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# Modeling and Predicting Momentum in Tennis Games

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## Abstract

As if being the first Wimbledon winner in over 20 years whose name wasn't Djokovic, Federer, Murray, or Nadal wasn't enough, Carlos Alcaraz's victory was even more astonishing considering he came back after being handily dealt with by Djokovic in the first set of the final 1-6. Alcaraz's miraculous recovery might be explained by a shift of flow in the game in his favor. In this paper, we compare two models that both interpret a small amount of past data within a given tennis match to determine which player has the current advantage, or "flow," in the next game. We then analyze the data provided to us, determine the advantages one player may have over another, and use them to further enhance our models. Afterward, we discuss our findings and how our model could be improved. We finish by generalizing our model to predict what would change if the tournament was held on a different playing surface or if it was used in Women's Professional Tennis.

# Introduction

Tennis is a unique sport in the context of data analytics. Singles matches are one-on-one, and few statistics have any immediate face value. As higher levels of technical aptitude are reached, with physical attributes beginning to plateau, mental fortitude becomes increasingly important. Any psychological effect capable of significantly influencing a game could provide a player with an advantage that has the potential to shift the outcome of a match. The phenomenon of momentum is a salient example of such a psychological effect. Despite being identified by players and coaches alike, arguments for the existence of momentum and its impact on the game of tennis are met with skepticism from mathematical psychologists, such as Amos Tversky (Hale).

## The Phenomenon of Momentum

Momentum, broadly speaking, is defined as “strength or force gained by motion or by a series of events” (“Momentum”). We are hesitant to use this definition, as concretely showing that the abstract concept of momentum exists is beyond the scope of our paper. Throughout the paper, we avoid giving a precise definition and, instead, investigate how we would expect it to affect the match being played, if the concept of momentum exists at all. Essentially, the aim of this paper is to see if local data describing the behavior of previous points can be used to predict the behavior of future points in a tennis match. If so, and we find a metric better than simply the number of points scored, we will be able to say that momentum at the very least likely exists.

## Assumptions

We made the conscious decision to focus on points scored rather than elapsed time for determining the distance between data points. This decision has the benefit of making the spacing between events uniform; however, this also eliminates the impact rest breaks may have on how the flow of the game changes, which could be an important factor in stopping the opponent’s momentum. We also assumed that the server has an inherent advantage, which we justify by finding the probability that the server of a game wins in our given data set (Data Dictionary.csv).

## The Data

The data we used were provided by the Consortium for Mathematics and its Applications (COMAP) as part of the 2024 Mathematical Contest in Modeling (MCM). This data included 31 matches from round three and forward of the Wimbledon Championship 2023 tournament. Of the 31 matches, we had to exclude two (Elahi Galan v. Ymer & Pella

v. Safiullin) as there was missing data from those matches that was required for our second model to function proficiently. We also chose to exclude the final three matches from any calculations made from the data set, as those games act as litmus tests to see how well our model predicts momentum outside of the given data set. All the data we had access to is in the appendix, which is the complete and unedited version from the Excel file we were given.

## Modeling Game Flow

### Notation

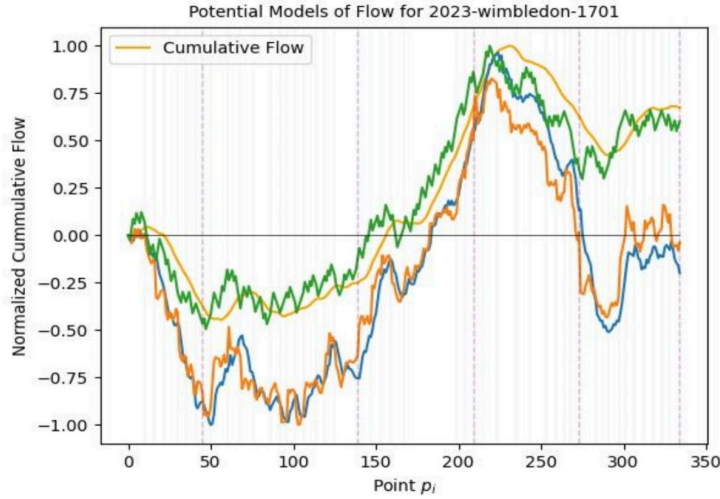
Before introducing our model for game flow, we will introduce the notation that will be used throughout the remainder of this paper. As is the convention in programming, we will use zero-based indexing.

It is often useful to be able to combine data values from a particular set of rows corresponding to a match, set, or game. In such instances, we will specify a collection of rows  $P = \{p_0, \dots, p_{|P|-1}\}$  where  $p_i$  is the  $i^{\text{th}}$  row in the data set. Particular rows within a data set will be referred to as points. The column data associated with the point  $p_i$  will be denoted as  $[\text{column name}]_i$ . Refer to the appendix for descriptions and values of column data.

If a point is the winning point of a game or a set, it may be referred to as a game point or a set point, respectively. A point is realized as a game or set point if its game victor or set victor data is non-zero.

For data set  $P$ , we define the window of width  $k$  at a point  $p_i$  to be the points  $\{p_{\max\{0, i-k\}}, \dots, p_i\}$ . Note that if  $i < k - 1$ , the first point in the window is  $p_0$ .

A quick note on the graphs in the paper: when the function is negative, then player 2 has the advantage, while the opposite is true if the function is positive. The dotted lines represent new sets, while the thin lines represent new games.



**Figure 1** There are many ways to describe the flow of a match. Each undefined line in the graph represents a different window size for the models used (see Table 1).

## Flow and Cumulative Flow

To begin analyzing the idea of momentum, we must define a metric that we can use to showcase its effect. We propose a model for flow, which will describe the overall trajectory of a game. Flow will mainly be defined by the net points scored by a player, namely, the difference between how many points player 1 scored and how many points player 2 scored. According to Craig O'Shannessy, Novak Djokovic's former statistician, only six players scored between 52 and 55 percent of the points they played between 2015 and 2019 (Farthing). These six players won 18 of the 20 grand slam tournaments held in the five-year period. So, if the point differential between two opponents is small, what must really matter is *when* they score in the game. Thus, a cumulative points model, indexed against a per-point step, will provide an effective base for a model of game flow.

The models in this paper are focused on identifying the net performance of the two players ( $p1\_points\_won - p2\_points\_won$ ). So, our convention is that positive measures are in favor of player one, and negative measures are in favor of player two.

To describe the flow of a match, we must first define the game flow step at the  $i^{\text{th}}$  point in a match. So, considering the advantage that the server has, we define the flow step at  $p_i$  to be

$$f(i) = (-1)^{point\_vector_i - 1} - \alpha(-1)^{server_i - 1} \quad (1)$$

where  $\alpha = 0.1835$  is the serving player's advantage as 0.5 minus the probability that the server wins the point. This probability was computed using the data in COMAP's "Wimbledon Featured Matches.Csv." An example of a computation using the flow step is as follows: if player one is the server and scores at  $p_i$ , then  $point\_vector = 1$  and  $server = 1$ , so  $f(i) = (-1)^0 - (0.1835)(-1)^0 = 1 - 0.1835 = 0.8165$ . If player one is the server and player two scores, then  $point\_vector = 0$  and  $server = 1$ , so  $f(i) = (-1)^1 - (0.1835)(-1)^0 = -1 - 0.1835 = -1.1835$ .

$victor = 2$  and  $server = 1$ , so  $f(i) = (-1)^1 - (-0.1835)(-1)^0 = -1 - 0.1835 = -1.1835$ . From this, we can see how server advantage is accounted for within flow steps. The flow step is the foundation of our match flow model, yet it provides little information about the net relative performance of the players. To further develop our model of the flow of a match, we define another metric that measures the relative performance of players within a window of points.

The  $k$ -flow step at  $p_i$  is the sum of flow steps over the window of width  $k$  about the point  $p_i$  and is denoted  $f_{(k)}(i)$ :

$$f_{(k)}(i) = \sum_{j=\max\{0, i-k+1\}}^i f(i) \quad (2)$$

The  $k$ -flow step at a point  $p_i$  can be interpreted as a measure of net performance in the last  $k$  points. Due to the  $k$ -flow step's ability to measure local net performance, we recognize it as a model and call it the KFS model.

The cumulative  $k$ -flow at  $p_i$  is the sum of  $k$ -flow steps over  $\{p_0, \dots, p_i\}$  and is denoted  $F_k(i)$ :

$$F_{(k)}(i) = \sum_{j=0}^i f_{(k)}(j) \quad (3)$$

We define the flow of a match, up to a given point in the game  $k$ , to be the cumulative  $k$ -flow. To choose an appropriate value of  $k$  we plotted  $k$  functions over different  $k$ -values until the function was approximately smooth as seen in Figure 1. For the rest of this paper, the flow of a match at a point  $p_i$  is defined to be the cumulative 18-flow and denoted simply as  $F(i)$ . This is our baseline global metric for the flow of a match. For two points of a match  $p_i$  and  $p_j$  where  $j > i$ , the change of flow between  $p_i$  and  $p_j$  is defined as

$$\frac{F(j) - F(i)}{j - i} \quad (4)$$

where  $\text{sgn}[\text{absolute value ok}](F(j) - F(i))$  is the direction of match flow between the points.

## Relationship Between Momentum and Flow

If the phenomenon of momentum exists and has an impact on player performance, then shifts in the net momentum of the players should affect the flow of the match. A shift in net momentum then manifests during points when the flow of the match shifts. If net momentum shifts in the favor of player one, then it would result in the flow of the game shifting to their favor. If momentum exists in an impactful way, then changes in net momentum can be detected by the local measure of performance that has an influence on global trends.

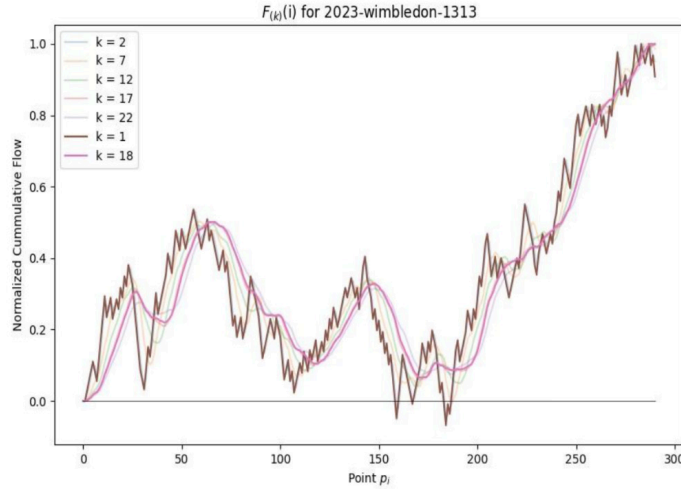


Figure 2 Cumulative  $k$ -flow for  $k = 1, 2, 7, 12, 17, 18, 22$ .

## Scaling Momentum Model

The game of tennis is composed of five scales of play: matches, sets, games, points, and rallies. Each scale of play is composed of plays of lower scales. Matches are composed of sets, sets of games, games of points, and points of rallies. Accounting for player performance at different scales of play provides a more comprehensive picture of the game state. In this section, we outline the definition of a model to predict net momentum that accounts for the scales of play seen within the game of tennis. We will refer to the model described below as the Scaling Momentum model (SM model). Note that before, we defined “local” as a window of width  $k$ , much less than the total points scored in a match,  $|P|$ . Now, we can more formally describe  $k \ll |P|$  to mean that the width of the window is at most on the scale of games.

We describe our measure of performance at the point scale the same way as we do the flow, except we divide by shots. For the purpose of this paper, though, we will be calling the number of shots in a point the rally count, as that is the language that was used in the instructions for us during the MCM competition. This heuristic measure is based on the idea that two evenly matched players would keep playing forever once the server advantage was overcome. So, given a point  $p_i$ , dividing  $(-1)^{\text{point\_victor}-1}$  by  $\text{rally\_count}_i$  provides a measure for how the players are performing in comparison to each other. We can extend this heuristic argument to get a measure of net performance on the scale of games. Instead of dividing by rallies within a game, we divide by the number of points scored in a game and multiply by the minimum number of points to win a game.

Furthermore, we can account for advantages on different scales of play. In the definition of our flow steps, we account for server advantage for a given point  $\alpha = 0.1835$ , but this is not the only advantage that is seen throughout the tennis game. On the scale of games, the server

of a game is much more likely to win the game. The probability of the server winning the game was calculated using the Wimbledon data provided to be approximately 0.85458. So, the advantage of the server for a given game is  $\gamma = 0.35458$ .

In the following model, we combine the heuristic measures of net performance and account for advantages on the scales of points and games. The goal of the SM model is to integrate information that the KFS model does not account for in order to search for potentially unidentified factors.

Given a data set  $P$  let  $G$  be the index set of game points in  $P$  and let  $Wk(i)$  denote the index set of a window of width  $k$  at point  $p_i$  in  $P$ . We define the  $k$  point moments as follows:

$$m_{p,(k)}(i) = \sum_{j \in W_k(i)} \frac{(-1)^{\text{point\_victor}_j - 1}}{\text{rally\_count}_j} - \alpha(-1)^{\text{server}_j - 1} \quad (5)$$

For  $g_i, g_{i+1} \in G$  define  $|g_i|$  to be the number of points won in the game point  $p_{g_i}$ . We can then define the  $k$  game moments as follows:

$$m_{g,(k)}(i) = 4 \sum_{g_j, g_{j-1} \in G \cap W_k(i)} \frac{(-1)^{\text{game\_victor}_{g_j} - 1}}{|g_j - g_{j-1}|} - \gamma(-1)^{\text{game\_victor}_{g_j} - 1} \quad (6)$$

It is important to note that the sum above requires both  $g_i$  and  $g_{i-1}$  to be in  $G$ . We scale by a factor of 4 to account for the difference in scales and to ensure that game moments and point moments both have the same units. In both the point and game moments, we ensure that the units are appropriate by scaling by the minimum number of rallies to win a point or points to win a game. To combine the  $k$  game and point moments in equations (5) and (6), we simply average the two values:

$$m_{(k)}(i) = \frac{m_{p,(k)}(i) + m_{g,(k)}(i)}{2} \quad (7)$$

The final SM model measures the relative performance of the players, accounting for advantages on the scale of points and games.

Just as with our definition of flow, we define  $M_{(k)}(i)$  to be a predicted cumulative measure of performance from the SM model. It may be referred to as a secondary game flow model:

$$M_{(k)}(i) = \sum_{j=0}^i m_{(k)}(j) \quad (8)$$

We will use the cumulative SM model in a comparative analysis of the efficiency of the SM model and the KFS model.

## Predicting Swings in Game Flow

Given a data set  $P$  where  $|P| = i$ , we say that a model provides a local measure of performance at  $p_j$  where  $j \leq i$  if it only depends on data within a window of width  $k \ll i$ , where  $k \ll i$  means that  $k$  is much less than  $i$ . In the section discussing scales of play, we note a more formal meaning of  $k \ll i$ . If all local measures of performance are ineffective at predicting the direction of the match flow, we can conclude that momentum likely does not exist in an impactful way. Conversely, if an accurate measure of local performance predicts the direction of the match flow, it is likely that momentum does exist and has an impact on the flow of the game.

Given a data set  $P$ , assume we know the data points  $p_0, \dots, p_i$ ; we can think of  $p_i$  as the current point. We can use the KFS model from (2) as a measure of local performance at  $p_j$ . We define that  $\mu_j = F_{(k)}(j)$ . We refer to this sum as the  $k$ -moments computed by a model. The  $k$ -moments identify which player is performing best at point  $p_j$ . If  $\mu_j > 0$ , then player one is performing better; if  $\mu_j < 0$ , then player two is performing better; and if  $\mu_j = 0$ , then the players are performing equally at  $p_j$ . Then, let  $\Delta_{\text{sgn}}(\mu)$  be the set of indices where the  $k$ -moments change signs; so, for  $j \in \Delta_{\text{sgn}}(\mu)$ , the sign of  $\mu_j$  is different than the sign of  $\mu_{j+1}$ .

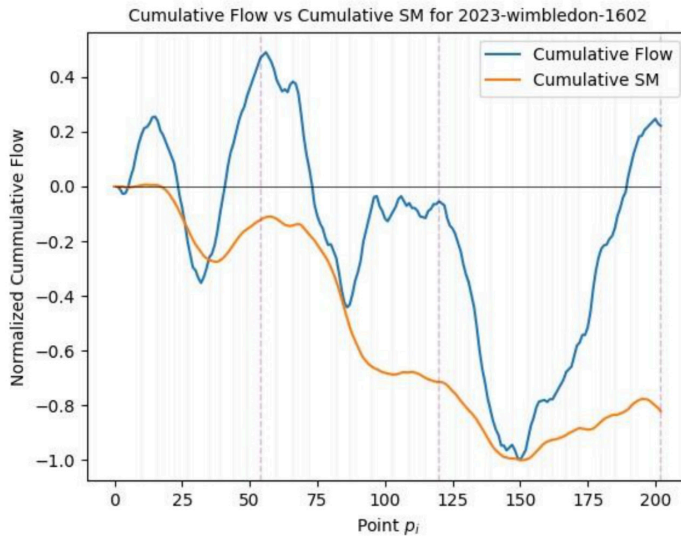


Figure 3 SM model flow change predictions vs. cumulative for Sinner v. Djokovic.

Given that changes in net momentum can be identified by a notable shift in a local measure of player performance, the points indexed by  $\Delta_{\text{sgn}}(\mu)$  are candidates for changes in net momentum and consequently could be indicators of a future change in the direction of match flow. The change in match direction predicted by the KFS model is computed as  $\text{sgn}(\mu_i - \mu_{i_{\text{last}}}) i_{\text{last}} = \max\{\Delta_{\text{sgn}}(\mu)\}$ .

This method for making flow change predictions based on a model can be repeated for the SM model as such: given a data set  $P$ , assume we know the points  $p_0, \dots, p_i$ ; we can think of  $p_i$  as the current point. We can use the SM model as a measure of local performance at  $p_i$ . We define that  $\mu_j = \{m_{(k)}(j)\}_{j=0}^i$ . The points indexed by  $\Delta \text{sgn}(\mu)$  are candidates for changes in net momentum. The change in match direction predicted by the SM model is computed as  $\text{sgn}(\mu_i - \mu_{i_{last}}) i_{last} = \max\{\Delta \text{sgn}(\mu)\}$ .

## Computing Accuracy of Predictions

The calculation of game flow is dependent on the  $k$ -flow steps, so the predictions made by the  $k$ -flow steps are reiterating information already known to the modeling of flow. Observe that the  $k$ -flow steps are a discrete difference of the cumulative  $k$ -flow steps. In our prior predictions we used the  $k$ -flow steps in a window to predict the flow of the match. However, these predictions can't claim to demonstrate the existence of hidden information because the values the KFS model is predicting depend solely on the data it is using. In order to reveal the possibility of hidden information, we must find a model that outperforms the KFS model while utilizing information that is not captured by the KFS model.

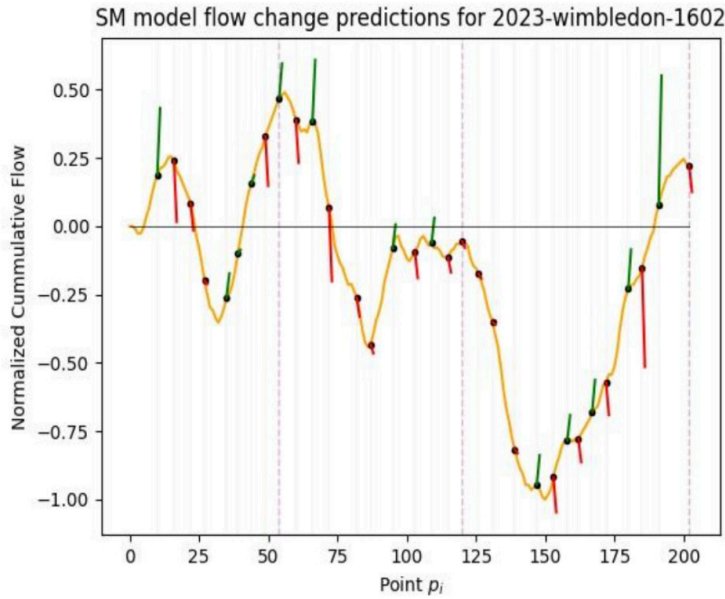


Figure 4 KFS model flow change predictions for Sinner v. Djokovic.

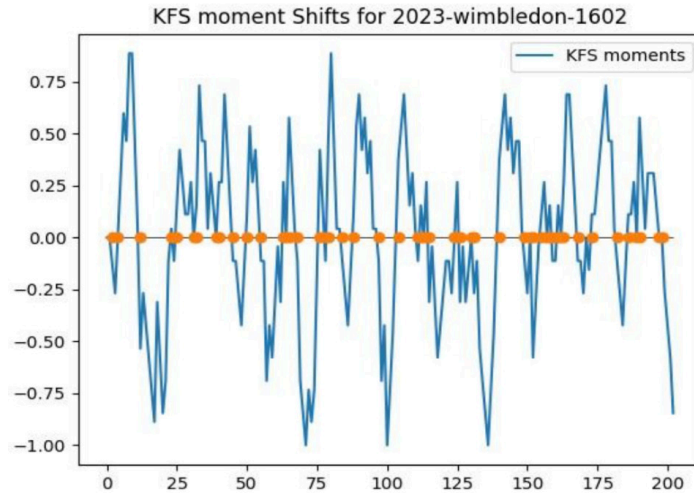


Figure 5 KFS model momentum shifts for Sinner v. Djokovic.

We compare the effectiveness of the KFS model to the effectiveness of the SM model. The scaling momentum model factors in more information than the KFS model. If the scaling SM model can outperform the KFS model at predicting the flow of the match, this would strongly suggest that the SM model is exploiting hidden factors such as momentum. Wanting to test the accuracy of our model, we computed a prediction for the direction change in flow at the end of each game. Then we compared the direction predicted by the model to the direction of the flow between the next game's points. We then took the average of the correct guesses across all game point predictions.

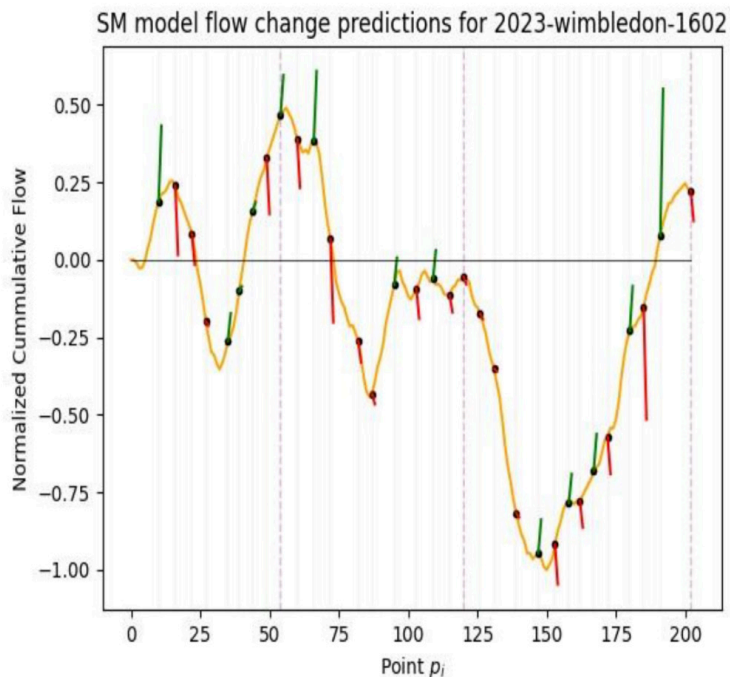


Figure 6 SM model flow change predictions for the Wimbledon semifinal match of Sinner v. Djokovic.

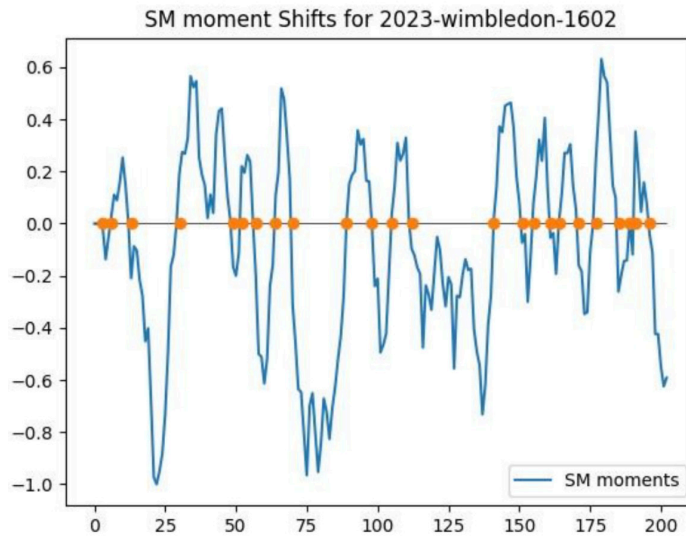


Figure 7 SM model momentum shifts for Sinner v. Djokovic.

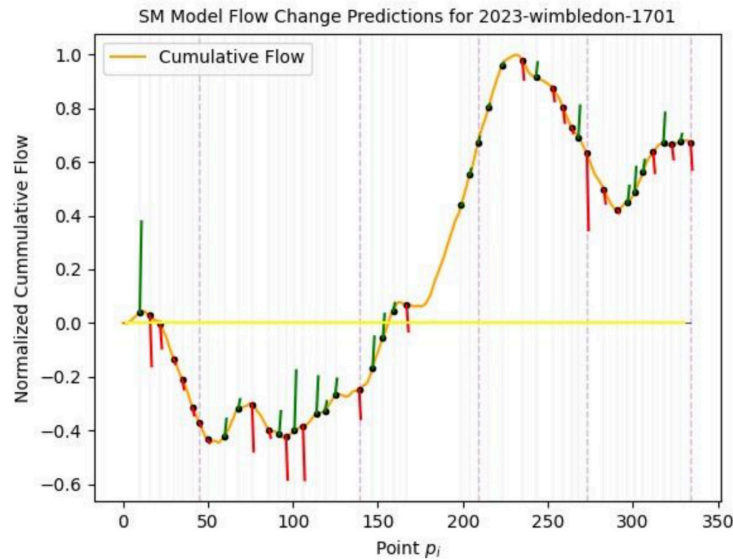
## Model Comparisons

Table 1 describes the accuracy of the models. Accuracy was calculated by taking the sign of the flow prediction and comparing it to the behavior of the graph of cumulative flow at the point of the next prediction.

The sign of flow change is how we measure the accuracy of our flow swing predictions. At the end of each game, we make a prediction of flow change based on the momentum of the game at the current point and the momentum at the last momentum change. The window for the prediction models has a notable impact on the relative accuracies if the window is exactly the length of the previous game. Since the game point from the prior game is used to compute the point count of the game point we are predicting from, the SM model's accuracy is negatively impacted because it cannot compute the momentum change from the most recent game. Altering the SM model to count beyond the window would increase the range of data that the SM model has and thus give it an unfair advantage over the KFS model.

Table 1 Model accuracy chart.

| Model Accuracy Chart                |           |       |          |       |
|-------------------------------------|-----------|-------|----------|-------|
|                                     | KFS Model |       | SM Model |       |
| Compared To Cumulative              | KFS       | SM    | KFS      | SM    |
| Static Window, $k=6$                | 65.0%     | 65.5% | 70.0%    | 75.0% |
| Dynamic Window, full game           | 68.0%     | 68.0% | 63.0%    | 70.0% |
| Dynamic Window, running average ppg | 65.0%     | 65.4% | 69.9%    | 74.5% |



**Figure 8** SM model flow change predictions for the 2023 Wimbledon final, Alcaraz v. Djokovic. This shows the shift in momentum in favor of Alcaraz in the middle of the match.

## Model Analysis

### Does Momentum Exist?

Our model seems to suggest that there is an advantage to winning points at certain moments, beyond what one would expect from a rudimentary model. This would indicate that a notion of momentum, as defined above, does exist.

### Our Shortcomings and Feats

We would have liked to have included the ability for our model to predict who the overall match winner would be, as currently, the future accuracy window is a small one, but having the future window width be relatively small was a necessary condition for our SM model to work properly, so we made the decision not to expand it. Despite this, we believe our model answers the principal question being asked: are there any external factors beyond normal play that indicate the swings that happen during a tennis match?

### Generalizing our Model

If we generalize our second model to include different types of tennis matches (e.g., different surface types) and we include Women's Professional Tennis, Lisi et al. seem to suggest that our model would still perform well. Because our model considers rally count an important factor in determining the flow-state of the game, if the rally count for different surfaces is similar to that of grass, we would expect our model to perform just as well. The researchers calculated the mean rally length on hard surface to be 1.1875 times that of grass,

with clay being 1.34375 times that of grass. Since the mean rally lengths for these surfaces are not just equal to but greater than they are for grass, we speculate that the longer the rally count, the more that flow would contribute to our model predicting a larger difference. Lisi et al. also shows that Women's Professional Tennis has longer rally counts on average, which would again suggest our model would predict that flow would play a larger part in those games. So, we predict our model will perform just as well, if not better than it currently performs in a wide range of tennis matches. While very difficult to predict, our SM model would theoretically function if used in a game like table tennis as long as it was updated to account for the longer games and the updated server advantage coefficient. However, unlike predicting model accuracy when applied to different surfaces and leagues, it is much more difficult to predict whether the model will work when applied to table tennis.

## Enhancing our Model

We could expand upon the second model by having whole sets affect the momentum of a match, a more comprehensive version of what our current model does right now. We could also improve upon it by having more factors affect the weight of each point, as right now, only three more metrics are taken into account when compared with the KFS model.

Extending the SM model to the scale of sets may provide more detail regarding the underlying momentum gained from winning or losing a set. This idea can be extended further to the scale of sets, matches, and rounds in a tournament. However, since our goal is to identify changes in flow on a local scale within a match, we restricted our definition of the SM model to the scale of games. If we were to extend the SM model to larger scales of play, we could then calculate the advantages at those scales as well. Furthermore, with a more comprehensive understanding of the inherent advantages of a given game state (such as player one being a server) independent of player skills and prior game states, we could further refine the SM model to be more accurate in identifying shifts in player performance.

## Acknowledgments

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# Appendix

**Table 2** The “Data Dictionary” given by COMAP.

| Variables     | Explanation                                                                 | Example                                                                         |
|---------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| match_id      | match identification                                                        | 2023-wimbledon-1701 (“7” is the round, and “01” the match number in that round) |
| player1       | first and last name of the first player                                     | Carlos Alcaraz                                                                  |
| player2       | first and last name of the second player                                    | Novak Djokovic                                                                  |
| elapsed_time  | time elapsed since start of first point to start of current point (H:MM:SS) | 0:10:27                                                                         |
| set_no        | set number in match                                                         | 1, 2, 3, 4, or 5                                                                |
| game_no       | game number in set                                                          | 1, 2, ...,7                                                                     |
| point_no      | point number in game                                                        | 1, 2, 3... etc.                                                                 |
| p1_sets       | sets won by player 1                                                        | 0, 1, or 2                                                                      |
| p2_sets       | sets won by player 2                                                        | 0, 1, or 2                                                                      |
| p1_games      | games won by player 1 in current set                                        | 0, 1,...,6                                                                      |
| p2_games      | games won by player 2 in current set                                        | 0, 1,...,6                                                                      |
| p1_score      | player 1’s score within current game                                        | 0 (love), 15, 30, 40, AD (advantage)                                            |
| p2_score      | player 2’s score within current game                                        | 0 (love), 15, 30, 40, AD (advantage)                                            |
| server        | server of the point                                                         | 1: player 1, 2: player 2                                                        |
| serve_no      | first or second serve                                                       | 1: first serve, 2: second serve                                                 |
| point_victor  | winner of the point                                                         | 1 if player 1 wins, 2 if player 2 wins                                          |
| p1_points_won | number of points won by player 1 in match                                   | 0, 1, 2... etc.                                                                 |
| p2_points_won | number of points won by player 2 in match                                   | 0, 1, 2... etc.                                                                 |
| game_victor   | a player won a game this point                                              | 0: no one, 1: player 1, 2: player 2                                             |
| set_victor    | a player won a set this point                                               | 0: no one, 1: player 1, 2: player 2                                             |
| p1_ace        | player 1 hit an untouchable winning serve                                   | 0 or 1                                                                          |
| p2_ace        | player 2 hit an untouchable winning serve                                   | 0 or 1                                                                          |
| p1_winner     | player 1 hit an untouchable winning shot                                    | 0 or 1                                                                          |
| p2_winner     | player 2 hit an untouchable winning shot                                    | 0 or 1                                                                          |

| <b>Variables</b>   | <b>Explanation</b>                                               | <b>Example</b>                                                        |
|--------------------|------------------------------------------------------------------|-----------------------------------------------------------------------|
| winner_shot_type   | category of untouchable shot                                     | F: Forehand,<br>B: Backhand                                           |
| p1_double_fault    | player 1 missed both serves and lost the point                   | 0 or 1                                                                |
| p2_double_fault    | player 2 missed both serves and lost the point                   | 0 or 1                                                                |
| p1_unf_err         | player 1 made an unforced error                                  | 0 or 1                                                                |
| p2_unf_err         | player 2 made an unforced error                                  | 0 or 1                                                                |
| p1_net_pt          | player 1 made it to the net                                      | 0 or 1                                                                |
| p2_net_pt          | player 2 made it to the net                                      | 0 or 1                                                                |
| p1_net_pt_won      | player 1 won the point while at the net                          | 0 or 1                                                                |
| p2_net_pt_won      | player 2 won the point while at the net                          | 0 or 1                                                                |
| p1_break_pt        | player 1 has an opportunity to win a game player 2 is serving    | 0 or 1                                                                |
| p2_break_pt        | player 2 has an opportunity to win a game player 1 is serving    | 0 or 1                                                                |
| p1_break_pt_won    | player 1 won the game player 2 is serving                        | 0 or 1                                                                |
| p2_break_pt_won    | player 2 won the game player 1 is serving                        | 0 or 1                                                                |
| p1_break_pt_missed | player 1 missed an opportunity to win a game player 2 is serving | 0 or 1                                                                |
| p2_break_pt_missed | player 2 missed an opportunity to win a game player 1 is serving | 0 or 1                                                                |
| p1_distance_run    | player 1's distance ran during point (meters)                    | 5.376, 21.384, etc.                                                   |
| p2_distance_run    | player 2's distance ran during point (meters)                    | 6.485, 12.473, etc.                                                   |
| rally_count        | number of shots during the point                                 | 1, 2, 4, etc. (includes serve)                                        |
| speed_mph          | speed of serve (miles per hour; mph)                             | 81, 124, etc.                                                         |
| serve_width        | direction of serve                                               | B: Body, BC: Body / Center,<br>BW:<br>Body / Wide, C: Center, W: Wide |
| serve_depth        | depth of serve                                                   | CTL: Close To Line,<br>NCTL: Not Close To Line                        |
| return_depth       | depth of return                                                  | D: Deep, ND: Not Deep                                                 |